

Practical No. 01

Title: Data Acquisition of Temperature Sensor

Aim: To raise an alarm signal when temperature exceeds a limiting value using Arduino Uno.

Apparatus: LM35 Temperature Sensor, Jumper Wires, Arduino Uno, Sound Buzzer, Computer, etc.

Theory:

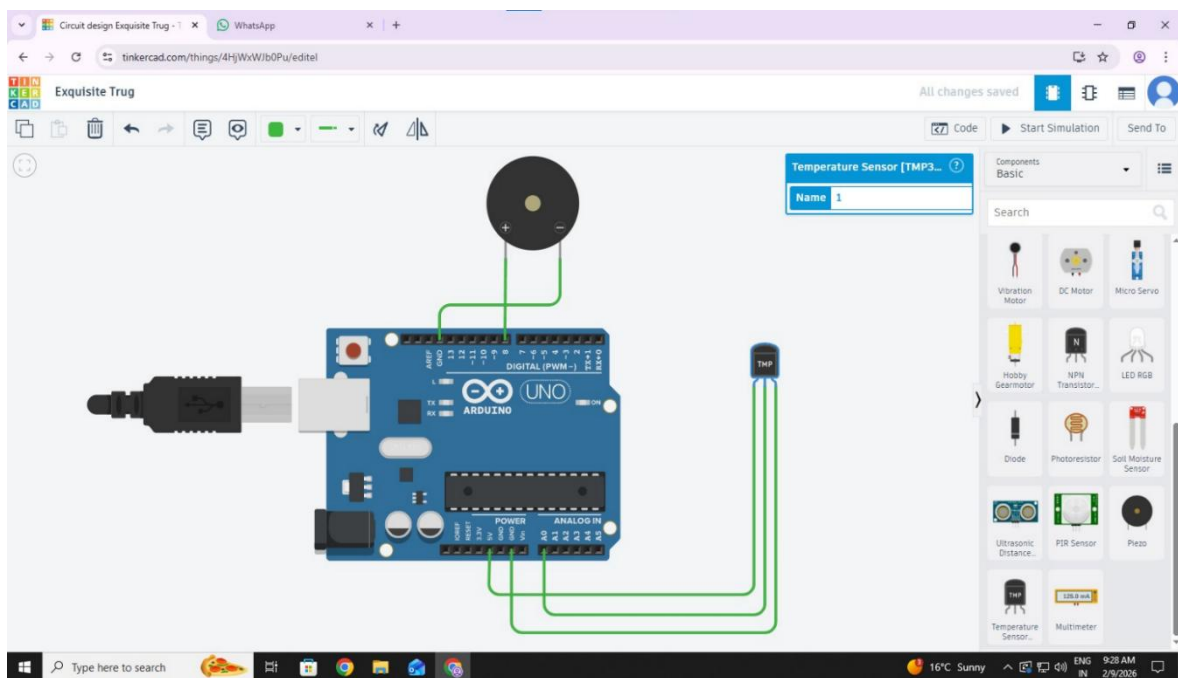
The LM35 Temperature Sensor senses temperature in its environment and convert it to corresponding voltage. Arduino Uno is a micro-controller used to process and read data from the Temperature Sensor. We set the limiting Temperature value to 50 Degree Celsius.

When the value exceeds 50, an actuating signal is generated in the Arduino Uno which then transfers the actuating signal to the Sound Buzzer.

Procedure:

1. Connect the power pin, signal pin, ground pin of LM35 Temperature sensor to 5V, A0 analog pin, GND of Arduino Uno.
2. Connect positive and negative pin of Sound Buzzer to the digital 8 pin and GND of Arduino Uno.
3. Connect Arduino and Computer using USB to mini USB cable.
4. Set the desire limiting Temperature Value.
5. Upload the C++ program using Arduino IDE to the PORT of Arduino.

Diagram:



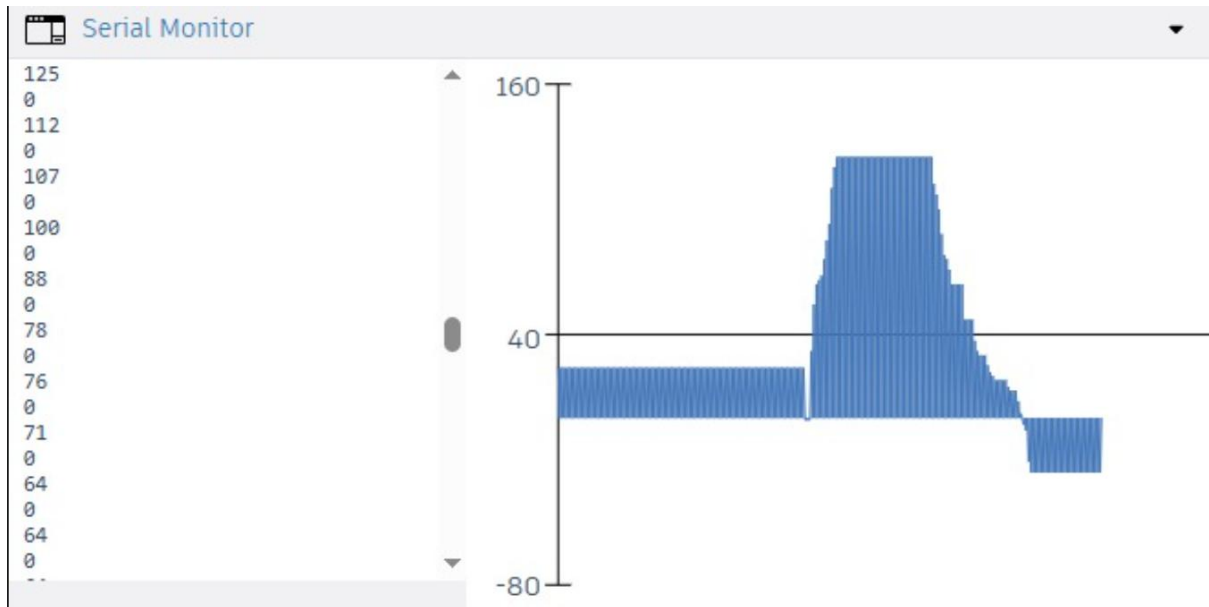
Program:

The program of Arduino is done using C++ language

```
int temp_voltage{0};  
int temp{0};  
int temp_pin{A0};  
int sound_pin{8};  
int timestamp{0};  
void setup()  
{  
  Serial.begin(9600);  
  pinMode(temp_pin, INPUT);  
  pinMode(sound_pin, OUTPUT);  
}  
void alert_sound()  
{  
  if(temp>50)  
  {  
    digitalWrite(sound_pin, HIGH);  
  }  
  else  
  {  
    digitalWrite(sound_pin, LOW);  
  }  
}  
void loop()  
{  
  temp_voltage=analogRead(temp_pin);  
  temp=map(temp_voltage, 104, 358, 0, 125);  
  Serial.println(temp);
```

```
Serial.println(timestamp);  
alert_sound();  
delay(100);  
}
```

Graph:



Conclusion:

Hence, we have finally learnt to produce an actuating signal from any sensor when value exceeds limiting value using Arduino.

Practical No. 2

Title: Interfacing data acquisition system hardware with computer to measure and control the robotic system.

Aim: To design and implement a system that can control the movement of a robotic arm using Arduino as a data acquisition system connected to a computer.

Apparatus: Arduino Uno, wires, Breadboard, USB cable, Robotic arm model, Computer, etc.

Theory:

A robotic arm is a mechanical arm that is designed to perform tasks that would otherwise be difficult or dangerous for a human to perform. Robotic arms are widely used in manufacturing, assembly, and other industries where precision and accuracy are critical. In order to control the position and movement of a robotic arm, it is necessary to use motors and sensors. In this experiment, we will use a servo motor connected to an Arduino board to control the position of a robotic arm with one joint.

A servo motor is a type of motor that is used to control the position of a device. Servo motors are commonly used in robotics, RC vehicles, and other applications that require precise control of movement. A servo motor consists of a small DC motor, a gear train, and a control circuit. The control circuit uses a feedback mechanism to adjust the position of the motor, and the gear train provides the necessary torque to move the load.

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are designed to make it easy to create interactive objects and environments. The boards are programmable and can be used to control motors, sensors, and other devices.

Controlling a Servo Motor with an Arduino:

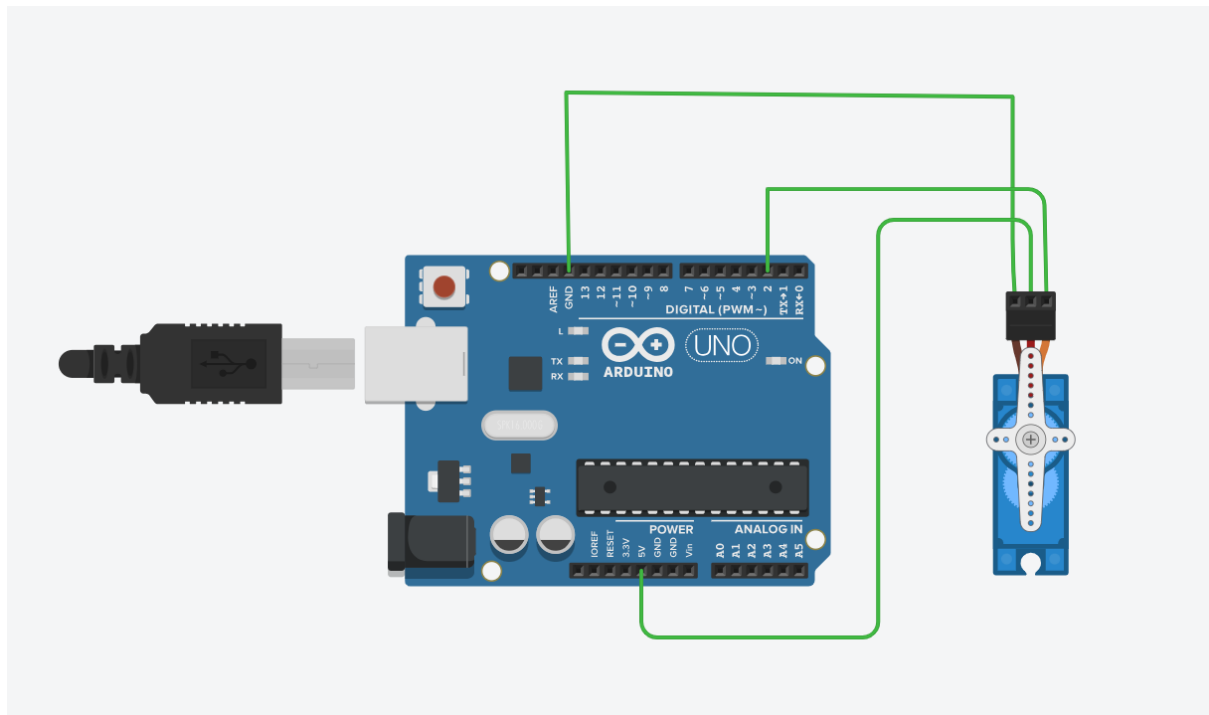
Controlling a servo motor with an Arduino is relatively simple. The servo motor is connected to one of the digital output pins on the Arduino board, and the Arduino sends a series of pulses to the servo motor to control its position. The duration of the pulse determines the position of the motor, and the Arduino can send different pulse durations to move the motor to different positions.

In this experiment, we will use the Servo library for Arduino to control the position of the servo motor. The servo library provides a simple way to control servo motors, and allows the user to specify the angle of the servo motor in degrees.

Controlling a Robotic Arm with a Servo Motor:

In order to control the position of a robotic arm with a servo motor, the motor must be connected to one of the joints of the arm. The angle of the joint can be controlled by sending a series of pulses to the servo motor, which in turn moves the joint to the specified position. By using multiple servo motors and sensors, it is possible to control the movement of the entire robotic arm.

Diagram:



Procedure:

1. Connect the servo motor to the Arduino board as follows:
2. Connect the signal pin of the servo motor to Pin 2 on the Arduino board.
3. Connect the VCC and GND pins of the servo motor to the 5V and GND pins of the Arduino board respectively.
4. Connect the robotic arm to the servo motor. Make sure that the servo motor is connected to the joint of the arm.
5. Upload the code to the Arduino board
6. Open the Serial Monitor in the Arduino IDE.
7. Enter the angle of the joint when prompted by the Serial Monitor. The angle should be in the range of 0 to 180.
8. Watch the robotic arm move to the specified position as you enter the angle in the Serial Monitor.

Program:

```
#include <Servo.h>

Servo servo_1;

int servo_pin (2);

int servo_1_angle (0);

int baud_rate (9600);

void setup() {

  Serial.begin(baud_rate);

  servo_1.attach(servo_pin);
```

```

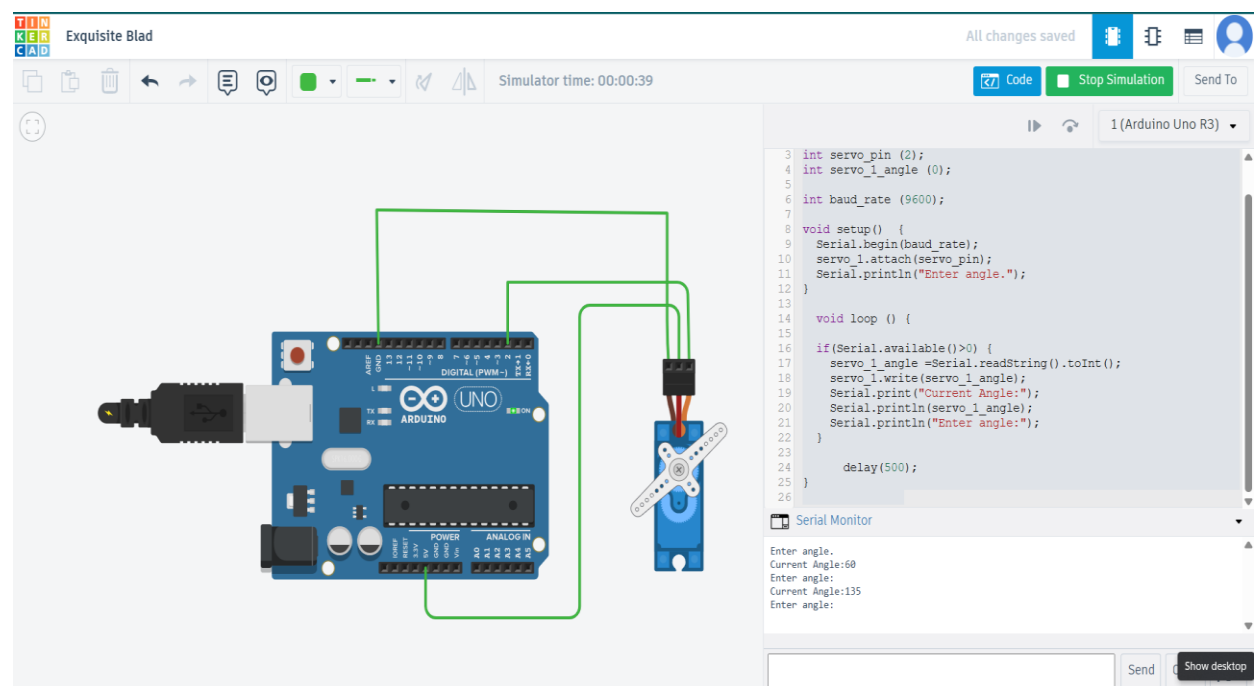
Serial.println("Enter angle.");
}

void loop () {
  if(Serial.available()>0) {
    servo_1_angle =Serial.readString().toInt();
    servo_1.write(servo_1_angle);
    Serial.print("Current Angle:");
    Serial.println(servo_1_angle);
    Serial.println("Enter angle:");
  }

  delay(500);
}

```

Serial Monitor:



Conclusion:

In conclusion, this experiment demonstrate the basic principles of robotic arm control using an Arduino and servo motor, and serves as a starting point for more complex projects involving jointed robotic arms. By adjusting the code and connecting additional motors and sensors, the user can create a more sophisticated system for controlling robotic arms.